P-S-311F
July 14, 1978
SUPERSEDING
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February 5, 1974

FEDERAL SPECIFICATION

SCOURING POWDER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers scouring powder for use in cleaning porcelain surfaces (vitreous types) and for general kitchen use (see 6.1).

1.2 Classification. The scouring powder covered by this specification shall be of the following types, as specified (see 6.2).

Type I - With bleaching agent.
Type II - Without bleaching agent.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-D-723 - Drums, Fiber.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standard:

D 820 - Chemical Analysis of Soaps Containing Synthetic Detergents.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Finished product. The finished product shall be a clean, uniform powder containing finely ground siliceous material, synthetic detergents, builders, and a bleaching compound (type I only). It shall be odorless or mildly perfumed. The finished product shall not contain lumps nor be caked up in its container.

3.1.1 Scouring powder shall be packaged in 14-ounce (397 g), 21-ounce (595 g), 1-pound (454 g), 5-pound (2.27 kg), 25-pound (11.3 kg), or 300-pound 136 kg) quantities, as specified (see 6.2).

3.2 Excluded materials. The scouring powder shall not contain ammonia releasing compounds, when tested as specified in 4.3.4.

3.3 Chemical and physical requirements. The scouring powder shall conform to the requirements of table I, when tested as specified in 4.3.1, 4.3.2 and 4.3.3.

TABLE I. Chemical and physical requirements

Property	Requirements			
	Type I		Type II	
	min.	max.	min.	max.
Matter volatile at 105°C (percent)	--	11.0	--	11.0
Anhydrous salt free synthetic detergent (percent)	1.7	7.0	1.7	7.0
Phosphates, as P_2O_5 (percent)	1.1	7.5	1.1	7.5
Insoluble siliceous matter (percent)	65.0	92.0	65.0	92.0
pH of 10% dispersion	10.5	12.0	10.5	12.0
Free alkali, as NaOH (percent)	--	0.1	--	0.1
Available chlorine (percent)	0.4	--		

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3.4 Fineness of insoluble siliceous matter. The insoluble siliceous matter shall comply with the fineness requirements specified in table II, when tested as specified in 4.3.5.

TABLE II. Fineness of insoluble siliceous matter

Retained on	Type I percent min.	Type I percent max.	Type II percent min.	Type II percent max.
No. 100 U.S. Standard Sieve	-	0.5	-	0.5
No. 200 U.S. Standard Sieve	-	10.0	-	10.0
No. 325 U.S. Standard Sieve	20.0	-	20.0	-

3.5 Labeling. The containers shall be marked or labeled with the manufacturer's normal directions for use.

3.6 Containers (type I). Type I scouring powder shall be supplied in commercial blow molded plastic containers or in spirally or convolutedly wound fiber cans with aluminum or plastic ends. The fiber cans shall have an outer ply of a paper-aluminum foil laminate. The interior surface of the fiber cans shall have a moisture resistant liner or coating of foil, cellulose acetate, polyethylene, wax, or resin. One end of each container shall be provided with punch-out or prepunched holes. Prepunched holes shall be covered with a moisture-resistant pressure sensitive tape closure.

3.7 EPA registration. When specified in the invitation for bids and contract for type I scouring powder, the scouring powder shall be registered with the Environmental Protection Agency as a disinfectant. The contractor shall have previously registered his scouring powder with the EPA and shall furnish the EPA registration number on his bid. The label shall contain the EPA registration number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery. Packaging, packing and marking shall be examined to determine compliance with the requirements of section 5. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, Acceptable Quality Level (AQL) 4.0, expressed in terms of percent defective. The sample unit shall be one shipping container fully prepared for delivery.

4.2.2 Examination of the scouring powder. The scouring powder shall be examined for compliance with the requirements of 3.1, 3.5, 3.6 and 3.7 (when applicable). Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, AQL 4.0 percent defective. The sample unit shall be one unit container (see 3.1.1).

4.2.3 Examination of net contents. Unit containers shall be examined for net contents. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, AQL of 2.5 percent defective. The net contents per container shall be not less than that specified in the contract.

4.2.4 Testing of the scouring powder. The scouring powder shall be tested as specified in 4.3 for compliance with the requirements of section 3. The lot shall be unacceptable if the sample fails any of the specified tests.

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4.3 Test methods.

4.3.1 Available chlorine. Weigh a 30-35 gram sample of the scouring powder to the nearest milligram. Transfer the sample to a 500 ml Erlenmeyer flask containing 250 ml of 3 percent potassium iodide solution. Add a Teflon coated stirring bar and place on a magnetic stirrer. Stir for 5 minutes. Add 15 ml of glacial acetic acid. (Caution: when the formulation contains a carbonate, the acid should be added a drop at a time to avoid an excessive rate of CO_2 evolution.) Add 15 ml starch indicator solution (see 4.3.1.1). The presence of available chlorine is indicated by the solution turning blue. Titrate with 0.1 N standardized sodium thiosulfate solution. Continue stirring throughout the titration. Continue titration until 1 drop of the thiosulfate solution causes the blue color to disappear. Record the volume of standard thiosulfate used and calculate the available chlorine as follows:

$$\text{Percent available chlorine} = \frac{\text{ml of thiosulfate} \times \text{N thiosulfate} \times 3.546}{\text{weight of sample}}$$

4.3.1.1 Starch indicator solution. Add 2 g of soluble starch and 10 mg of mercuric iodide to 1 liter of boiling distilled water. Continue boiling until all the starch has dissolved. Cool to room temperature and store in a glass-stoppered bottle. Before each use, test the starch indicator solution for deterioration by adding 5 ml of the starch indicator solution, 50 mg of KI and 1 drop of 0.1 N iodine solution to 100 ml distilled water. If the water turns blue, the starch indicator solution is suitable for use.

4.3.2 Analysis. Analyze the scouring powder for the following characteristics by the referenced sections in ASTM D 820.

4.3.2.1 Moisture and volatile matter at 105°C - sections 5-6.

4.3.2.2 Free alkali - sections 7-8.

4.3.2.3 Matter insoluble in water - section 15.

4.3.2.4 Phosphates - sections 21-28 (if P_2O_5 is greater than 2 percent), sections 29-34 (if P_2O_5 is 2 percent or less).

4.3.2.5 Synthetic detergent - sections 9-14, 35-41, and 46.

4.3.3 pH of 10 percent dispersion. Weigh out 30 + 0.1 grams of scouring powder and transfer to a 500 ml beaker. Add 270 ml of distilled or deionized water to the beaker and a Teflon coated stirring bar and place on a magnetic stirrer. Stir the mixture for 5 minutes. Determine the pH of the slurry with the pH meter and high alkalinity glass electrodes that have been standardized at pH 10.5 with standard buffer solution.

4.3.4 Ammonia releasing compounds. Place 0.2 gram of scouring powder in a small test tube. Add 2 ml of water and 2 drops of dilute NaOH solution. Immediately place a piece of moist red litmus paper in the mouth of the test tube and heat gently. Presence of ammonia releasing compounds is indicated by the litmus paper turning blue.

4.3.5 Fineness. Dry the No. 100, No. 200 and No. 325 U.S. Standard sieves in an oven at 105°C for 1 hour, cool to room temperature, and weigh to the nearest mg. Weigh a 100 g sample of the scouring powder to the nearest mg, transfer to a beaker, add 200 ml water and heat on a steam bath for 1 hour to dissolve the soluble matter. Nest the sieves, with the No. 100 sieve on top. Pour the material from the beaker through the sieves. Rinse the beaker with hot (85° - 95°C) water until all material has been transferred from the beaker to the sieves. Wash the material on the sieves with hot (85° - 95°C) water. If the material forms lumps or aggregates on washing, a camel's hair brush may be used to break up the lumps. Continue washing until no more material passes through the top sieve. Check by removing the top sieve and washing the sieve with 200 ml hot water, collecting the water in a 400 ml beaker. Stir the water in the beaker vigorously and place the beaker on a black surface. If particles are seen collected about the middle of the bottom of the beaker, wash all material from the beaker into the No. 200 sieve, and wash the No. 100 sieve with another

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200 ml of hot water as above. Continue washing the No. 100 sieve until no more particles are seen collected about the middle of the bottom of the beaker. Dry the No. 100 sieve in an oven at 105°C to constant weight, and weigh to the nearest mg. Determine the percent by weight of the original sample retained on the No. 100 sieve. Repeat the above procedure to determine the weight of the material retained on the No. 200 and No. 325 sieves. Calculate the percent of the insoluble material that does not pass through each of the sieves.

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Fourteen and twenty-one ounce cans. Scouring powder shall be packaged in one of the containers specified in 5.1.1.1.1 or 5.1.1.1.2. Unless otherwise specified, (see 6.2) the choice of container shall be at the option of the supplier. One end of each container shall be provided with a minimum of six punch-out holes or prepunched holes. The prepunched holes shall be covered with a moisture-resistant pressure sensitive adhesive tape closure.

5.1.1.1.1 A spirally or convolutely wound fiber can with aluminum or plastic ends may be used. For type I scouring powder, the outer ply shall be a paper-aluminum foil laminate. The can shall have an interior moisture resistant liner or coating of foil, cellulose acetate, polyethylene, wax or resin. For type II scouring powder, the can shall consist of an outer paper-aluminum foil laminate secured to a three ply spirally or convolutely wound fiber can.

5.1.1.1.2 A commercial type blow molded plastic container may also be used.

5.1.1.2 One-pound and five-pound boxes. One pound or 5 pounds of the scouring powder shall be packed in a folding paperboard water resistant box. The box shall be closed to prevent accidental opening.

5.1.1.3 Twenty-five pound box or sack. Twenty-five pounds of scouring powder shall be packaged in a top opening polyethylene bag. The length of the bag shall be a minimum of 4 inches greater than the filled bag. The polyethylene flexible film shall be 0.002 inch thick. The open end of the bag shall be folded over onto itself and secured with pressure-sensitive adhesive tape. The closed filled bag shall then be packaged in a fiberboard box conforming to PPP-B-636, type CF, class weather resistant. The box shall be closed in accordance with appendix of the box specification.

5.1.1.4 Three-hundred pound drum. Three-hundred pounds of scouring powder shall be packaged in a fiber drum conforming to PPP-D-723, type III, grade A. Drums shall be lined with a barrier or interior coating in accordance with normal practice.

5.1.2 Commercial. The scouring powder shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Fourteen-ounce, 21-ounce, 1-pound and 5-pound packages. Forty-eight 14-ounce cans, thirty 21-ounce cans, forty-eight 1-pound boxes or twelve 5-pound boxes of scouring powder, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, class weather-resistant grade V3c. The box shall be provided with top, bottom, and layer fiberboard pads. The box shall be closed, waterproofed and banded in accordance with the appendix to PPP-B-636. Alternatively, cleated plywood, wire bound or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape or adhesive.

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5.2.2 Commercial. The packaged scouring powder shall be packed in shipping containers to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The scouring powders covered by this specification are intended for use in the cleaning of porcelain surfaces (vitreous types). Type I scouring powder has bleach to aid in the removal of stains. The scouring powders covered by this specification should not be used on glass, stainless steel or other materials where the abrasive action of the silica grit is not desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) If EPA registration is required (see 3.7).
(d) Quantity required (see 3.1.1).
(e) Packaging and packing required (see 5.1 and 5.2).
(f) If a particular kind of 14 or 21-ounce container is required (see 5.1.1.1).
(g) Marking required (see 5.3).

MILITARY INTEREST:

Custodians:

Army - GL (MCA)
Air Force - 99

Reviewer:

Army - MD

User:

Navy - MC, MS, SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
EPA

Preparing activity:

GSA-FSS

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